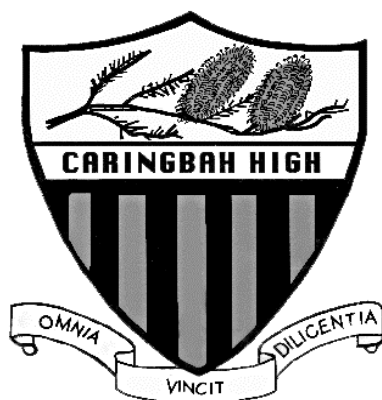




Caringbah High School

Year 12 2023

Mathematics Extension 2

HSC Course

Assessment Task 3

General Instructions

- Reading time – 5 minutes
- Working time – 75 minutes
- Write using a black pen
- NESA-approved calculators may be used
- A reference sheet is provided.
- In Questions 6-9, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 45

Section I 5 marks

Attempt Questions 1-5
Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 40 marks

Attempt Questions 6-9
Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only						
Section I	Section II				Total	
Q 1-5	Q6	Q7	Q8	Q9		
/5	/10	/10	/10	/10	/45	%

Section I

5 marks

Attempt Questions 1-5

Allow about 10 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5

1. What is the correct first step in integrating the following?

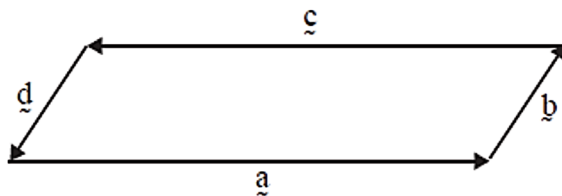
$$\int x e^x dx$$

- (A) $= x e^x - \int x dx$
- (B) $= x e^x - \int e^x dx$
- (C) $= x e^x - \int x e^x dx$
- (D) $= x e^x + \int x e^x dx$

2. The algebraic fraction $\frac{x+1}{2(x+h)^2}$, where h is a non-zero real number can be written in partial fraction form, where A and B are real numbers, as which of the following?

- (A) $\frac{A}{x+h} + \frac{B}{x+h}$
- (B) $\frac{A}{2x+h} + \frac{B}{(x+h)^2}$
- (C) $\frac{A}{x+h} + \frac{B}{(x+h)^2}$
- (D) $\frac{A}{2(x+h)} + \frac{B}{x+h}$

3. In the parallelogram $|\vec{a}| = 2|\vec{b}|$



Which one of the following statements is true ?

- (A) $\vec{a} = 2\vec{b}$
 - (B) $\vec{a} + \vec{b} = \vec{c} + \vec{d}$
 - (C) $\vec{a} + \vec{c} = 0$
 - (D) $\vec{a} - \vec{b} = \vec{c} - \vec{d}$
4. Of the following expressions which does NOT contain a logarithmic integral?

- (A) $\frac{x-1}{x^3+1}$
- (B) $\frac{x+1}{x^3-1}$
- (C) $\frac{x-1}{x^3-1}$
- (D) $\frac{x^2+1}{x^3+1}$

5. A, B and C are three collinear points with position vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ respectively. B lies between A and C .

If $2|\overrightarrow{BC}| = |\overrightarrow{AB}|$, the $\underline{c}$ is equal to which of the following?

(A) $\frac{3}{2}\underline{b} - \frac{1}{2}\underline{a}$

(B) $\frac{3}{2}\underline{a} - \frac{1}{2}\underline{b}$

(C) $\frac{1}{2}\underline{a} - \frac{3}{2}\underline{b}$

(D) $\frac{1}{2}\underline{b} - \frac{3}{2}\underline{a}$

End of Section I

Section II

40 marks

Attempt Questions 6–9

Allow about 65 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (10 marks) Use a SEPARATE writing booklet.

(a) Evaluate

3

$$\int_0^{\frac{\pi}{3}} \cos^3 3x \sin 3x \, dx$$

(b) The equation of intersecting line L and M are given below with respect to the origin O .

2

$$L: \quad \underset{\sim}{r} = \begin{bmatrix} 11 \\ 2 \\ 17 \end{bmatrix} + \lambda \begin{bmatrix} -2 \\ 1 \\ -4 \end{bmatrix}$$

$$M: \quad \underset{\sim}{r} = \begin{bmatrix} -5 \\ 11 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} p \\ 2 \\ 2 \end{bmatrix}$$

If L and M are perpendicular what is the value of p ?

(c) Evaluate the following

3

$$\int_0^1 \frac{x+1}{x^2+1} \, dx$$

(d) Given $\underset{\sim}{u} = 2\underset{\sim}{i} + 2\underset{\sim}{j} - 3\underset{\sim}{k}$ and $\underset{\sim}{v} = \underset{\sim}{i} + 2\underset{\sim}{j} + 2\underset{\sim}{k}$ find $\underset{\sim}{w}$ given:

2

$$\left| \underset{\sim}{w} \right|^2 = \left| \underset{\sim}{u} \right|^2 + \left| \underset{\sim}{v} \right|^2$$

End of Question 6

Question 7 (10 marks) Use a SEPARATE writing booklet.

(a) Evaluate

2

$$\int \frac{dx}{\sqrt{9 - 4x^2}}$$

(b) Use the substitution $u = 1 + x^2$ to evaluate

3

$$\int_0^{\sqrt{3}} \frac{x^3}{\sqrt{(1 + x^2)^3}} dx$$

(c) i) Find real numbers a and b such that

2

$$\frac{5x^2 - 3x + 1}{(x^2 + 1)(x - 2)} \equiv \frac{ax + 1}{x^2 + 1} + \frac{b}{x - 2}$$

ii) Find:

3

$$\int \frac{5x^2 - 3x + 1}{(x^2 + 1)(x - 2)} dx$$

End of Question 7

Question 8 (10 marks) Use a SEPARATE writing booklet.

(a) Evaluate:

3

$$\int \frac{1}{2x \ln 2x} dx$$

(b) Evaluate:

4

$$\int \sin^{-1} x \, dx$$

(c) Evaluate:

3

$$\int_{-2}^{-1} \frac{1}{\sqrt{-x^2 - 4x - 3}} dx$$

End of Question 8

Question 9 (10 marks) Use a SEPARATE writing booklet.

- (a) By using $t = \tan \frac{x}{2}$ find the integral

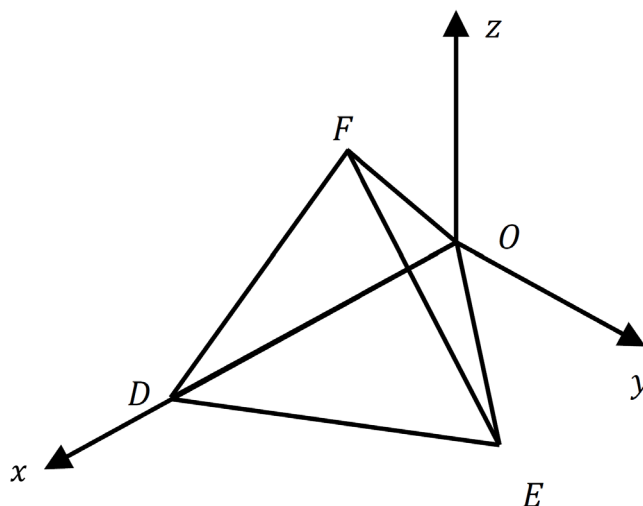
2

$$\int \frac{dx}{1 + \sin x}$$

- (b) Let $ODEF$ be a tetrahedron with faces that are equilateral triangles all with side lengths of 2 units.

4

Its base ODE lies flat on the xy -plane with two vertices at O and D , with F above the xy -plane.



Prove, using vectors, the coordinates of the vertex F are $\left(1, \frac{1}{\sqrt{3}}, \frac{2\sqrt{2}}{\sqrt{3}}\right)$

- (c) Consider the integral

4

$$I_n = \int_0^1 x^n e^{\frac{x^2}{2}} dx.$$

The recurrence relationship linking I_n and I_{n-2} can be written in the form

$$I_n = a - f(n) \times I_{n-2},$$

where n is an integer greater than or equal to 2, a is a constant and $f(n)$ is a function of n .

Find a and $f(n)$.

End of Examination

Multiple Choice Answer Sheet

Name _____

Colour the oval A, B, C or D that best answers the question.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct ↗

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

2023 YEAR 12 EXTENSION 2 AT3 EXAM SOLUTIONS

Q1. $xe^x - \int e^x dx$ (B)

Q2. $\frac{A}{x+h} + \frac{B}{(x+h)^2}$ (C)

The 2 can be absorbed into the values of A & B so not necessary for the denominator.

Q3. $\tilde{a} = -\tilde{c} \therefore \tilde{a} + \tilde{c} = 0$ (C)

Q4. $\frac{x-1}{x^3-1} = \frac{(x-1)}{(x-1)(x^2+x+1)} = \frac{1}{x^2+x+1}$ (C)

Q5. $\frac{3}{2}\tilde{b} - \frac{1}{2}\tilde{a}$ (A)

Q6. a) Let $u = \cos 3x \therefore du = -3 \sin 3x dx$

$$\int \cos^3 3x \sin 3x dx = -\frac{1}{3} \int u^3 du = -\frac{u^4}{12} + c \quad \text{①}$$

$$= -\left[\frac{\cos^4 3x}{12}\right]_0^{\frac{\pi}{3}} \quad \text{②}$$

$$= \frac{1}{12} \quad \text{③}$$

b) $\begin{bmatrix} -2 \\ 1 \\ -4 \end{bmatrix} \cdot \begin{bmatrix} p \\ 2 \\ 2 \end{bmatrix} = 0 \quad \text{①}$

$$\therefore -2p + 2 - 8 = 0 \therefore p = -3 \quad \text{②}$$

c) $\int_1^2 \frac{x+1}{x^2+1} dx = \int_1^2 \frac{x}{x^2+1} + \frac{1}{x^2+1} dx \quad \text{①}$

$$= \frac{1}{2} \int_1^2 \frac{2x}{x^2+1} dx + \int_1^2 \frac{1}{x^2+1} dx$$

$$= \frac{1}{2} \ln |x^2 + 1| + \tan^{-1} x \Big|_0^1 \quad \text{②}$$

$$= \frac{\ln 2}{2} + \frac{\pi}{4} \quad \text{③}$$

d) As $|\tilde{w}|^2 = |\tilde{u}|^2 + |\tilde{v}|^2$

therefore the vectors represent the sides of a right triangle. ①

$$\therefore \tilde{w} = \tilde{u} + \tilde{v} = (2\tilde{i} + 2\tilde{j} - 3\tilde{k}) + (\tilde{i} + 2\tilde{j} + 2\tilde{k})$$

$$= 3\tilde{i} + 4\tilde{j} - \tilde{k} \quad \text{②}$$

Q7. a) $\int \frac{dx}{\sqrt{9-4x^2}} = \frac{1}{2} \int \frac{dx}{\sqrt{\left(\frac{3}{2}\right)^2 - x^2}} \quad \text{①}$

$$= \frac{1}{2} \sin^{-1} \left(\frac{2x}{3} \right) + C \quad \text{②}$$

b) $u = 1 + x^2 \therefore du = 2x dx \therefore dx = \frac{1}{2x} du \quad \text{①}$

$$\int_0^{\sqrt{3}} \frac{x^3}{\sqrt{(1+x^2)^3}} dx = \frac{1}{2} \int \frac{u^{-1}}{u^{\frac{3}{2}}} du$$

$$= \frac{1}{2} \int u^{-\frac{1}{2}} - u^{-\frac{3}{2}} du$$

$$= \frac{1}{2} \left[2u^{\frac{1}{2}} + 2u^{-\frac{1}{2}} \right] \quad \text{②}$$

$$= \left[(1+x^2)^{\frac{1}{2}} + (1+x^2)^{-\frac{1}{2}} \right] \Big|_0^{\sqrt{3}} = \frac{1}{2} \quad \text{③}$$

c)i) $\frac{5x^2-3x+1}{(x^2+1)(x-2)} \equiv \frac{ax+1}{x^2+1} + \frac{b}{x-2}$

$$\therefore 5x^2 - 3x + 1$$

$$= (x-2)(ax+1) + b(x^2+1)$$

$$\therefore x = 2 \rightarrow 15 = 5b \therefore b = 3 \quad \text{①}$$

$$\therefore x = 1 \rightarrow 3 = -(a+1) + 6 \therefore a = 2 \quad \text{②}$$

c)ii) $\int \frac{5x^2-3x+1}{(x^2+1)(x-2)} dx = \int \frac{2x+1}{x^2+1} + \frac{3}{x-2} dx \quad \text{①}$

$$= \int \frac{2x}{x^2+1} + \frac{1}{x^2+1} + \frac{3}{x-2} dx \quad \text{②}$$

$$= \ln |x^2 + 1| + \tan^{-1} x + 3 \ln |x - 2| + C \quad \text{③}$$

Q8. a) Let $u = \ln 2x \therefore du = \frac{2}{2x} dx = \frac{1}{x} dx \quad \text{①}$

$$\therefore \int \frac{1}{2x \ln 2x} dx = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln |u| + C \quad \text{②}$$

$$= \frac{1}{2} \ln |\ln 2x| + C \quad \text{③}$$

b) $\int \sin^{-1} x dx$

$u = \sin^{-1} x$	$\frac{dv}{dx} = 1$
$\frac{du}{dx} = \frac{1}{\sqrt{1-x^2}}$	$v = x$

①

$$= x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx \quad \text{②}$$

$$= x \sin^{-1} x + \frac{1}{2} \int \frac{-2x}{\sqrt{1-x^2}} dx$$

$$\text{Let } v = 1 - x^2 \therefore dv = -2x dx$$

$$= x \sin^{-1} x + \frac{1}{2} \int \frac{1}{\sqrt{v}} dv \quad \text{③}$$

$$= x \sin^{-1} x + \frac{1}{2} [2\sqrt{v}] + C$$

$$= x \sin^{-1} x + \sqrt{1-x^2} + C \quad \text{④}$$

$$\begin{aligned} \text{Q8. c) } \int_{-2}^{-1} \frac{1}{\sqrt{-x^2-4x-3}} dx &= \int_{-2}^{-1} \frac{1}{\sqrt{1-(x+2)^2}} dx \quad \text{①} \\ &= \sin^{-1}(x+2) \Big|_{-2}^{-1} = \frac{\pi}{2} \quad \text{② ③} \end{aligned}$$

$$\begin{aligned} \text{Q9. a) } \int \frac{dx}{1+\sin x} \therefore \sin x &= \frac{2t}{1+t^2} \quad dx = \frac{2}{1+t^2} dt \\ &= \int \frac{1}{1+\frac{2t}{1+t^2}} \times \frac{2}{1+t^2} dt \\ &= 2 \int \frac{1}{(1+t)^2} dt \quad \text{①} \\ &= \frac{-2}{1+t} = \frac{-2}{1+\tan \frac{x}{2}} \quad \text{②} \end{aligned}$$

b) $\angle FOD = \angle FOE = \frac{\pi}{3}$ (equilateral triangle)

In ΔFOD :

$$\cos \angle FOD = \frac{\vec{OF} \cdot \vec{OD}}{|\vec{OF}| |\vec{OD}|}$$

$$\therefore \cos \frac{\pi}{3} = \frac{\begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}}{2 \times 2}$$

$$\therefore \frac{1}{2} = \frac{2a + 0 + 0}{4}$$

$$\therefore a = 1 \quad \text{①}$$

We have $E(1, \sqrt{3}, 0)$ using exact triangles

In ΔFOE :

$$\cos \angle FOE = \frac{\vec{OF} \cdot \vec{OE}}{|\vec{OF}| |\vec{OE}|}$$

$$\therefore \cos \frac{\pi}{3} = \frac{\begin{bmatrix} 1 \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} 1 \\ \sqrt{3} \\ 0 \end{bmatrix}}{2 \times 2}$$

$$\therefore \frac{1}{2} = \frac{1 + b\sqrt{3} + 0}{4}$$

$$\therefore b = \frac{1}{\sqrt{3}} \quad \text{②}$$

$$\text{As } |\vec{OF}| = 2 \quad \therefore \sqrt{(1)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 + c^2} = 2$$

$$\therefore c = \frac{2\sqrt{2}}{\sqrt{3}} \quad \text{③}$$

$$F \text{ is } \left(1, \frac{1}{\sqrt{3}}, \frac{2\sqrt{2}}{\sqrt{3}}\right) \quad \text{④}$$

$$\text{c) } I_n =$$

$$\int_0^1 x^n e^{\frac{x^2}{2}} dx \quad \boxed{\begin{array}{ll} u = \frac{1}{2}x^{n-1} & \frac{dv}{dx} = x e^{\frac{x^2}{2}} \\ \frac{du}{dx} = \frac{n-1}{2}x^{n-2} & v = e^{\frac{x^2}{2}} \end{array}} \quad \text{①}$$

$$= \frac{1}{2} \left[x^{n-1} e^{\frac{x^2}{2}} \right]_0^1 - \frac{n-1}{2} \int_0^1 x^{n-2} e^{\frac{x^2}{2}} dx \quad \text{②}$$

$$= \frac{1}{2} \left(e^{\frac{1}{2}} - 0 \right) - \frac{n-1}{2} \int_0^1 x^{n-2} e^{\frac{x^2}{2}} dx$$

$$= \frac{e^{\frac{1}{2}}}{2} - \frac{n-1}{2} \times I_{n-2} \quad \text{③}$$

$$\therefore a = \frac{e^{\frac{1}{2}}}{2}, f(n) = \frac{n-1}{2} \quad \text{④}$$